

Q1.

- (a) Hooke's law states that the tension in a stretched string of natural length l and modulus of elasticity λ is $\frac{\lambda x}{l}$ when its extension is x .

Using this formula, prove that the work done in stretching a string from an unstretched position to a position in which its extension is e is $\frac{\lambda e^2}{2l}$.

(3)

- (b) A particle, of mass 5 kg, is attached to one end of a light elastic string of natural length 0.6 metres and modulus of elasticity 150 N. The other end of the string is fixed to a point O .

- (i) Find the extension of the elastic string when the particle hangs in equilibrium directly below O .

(2)

- (ii) The particle is pulled down and held at the point P , which is 2 metres vertically below O .

Show that the elastic potential energy of the string when the particle is in this position is 245 J.

(2)

- (iii) The particle is released from rest at the point P . Find the speed of the particle when it reaches O .

(4)

(Total 11 marks)

Q2.

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- (a) Hooke's law states that the tension in a stretched string of natural length l and modulus of elasticity λ is $\frac{\lambda x}{l}$ when its extension is x .

Using this formula, prove that the work done in stretching a string from an unstretched position to a position in which its extension is e is $\frac{\lambda e^2}{2l}$.

(3)

- (b) A particle, of mass 5 kg, is attached to one end of a light elastic string of natural length 0.6 metres and modulus of elasticity 150 N. The other end of the string is fixed to a point O .

- (i) Find the extension of the elastic string when the particle hangs in equilibrium directly below O .

(2)

- (ii) The particle is pulled down and held at the point P , which is 0.9 metres

vertically below O .

Show that the elastic potential energy of the string when the particle is in this position is 11.25 J.

(2)

- (iii) The particle is released from rest at the point P . In the subsequent motion, the particle has speed $v \text{ m s}^{-1}$ when it is x metres **above** P .

Show that, while the string is taut,

$$v^2 = 10.4x - 50x^2$$

(7)

- (iv) Find the value of x when the particle comes to rest for the first time after being released, given that the string is still taut.

(2)

(Total 16 marks)



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